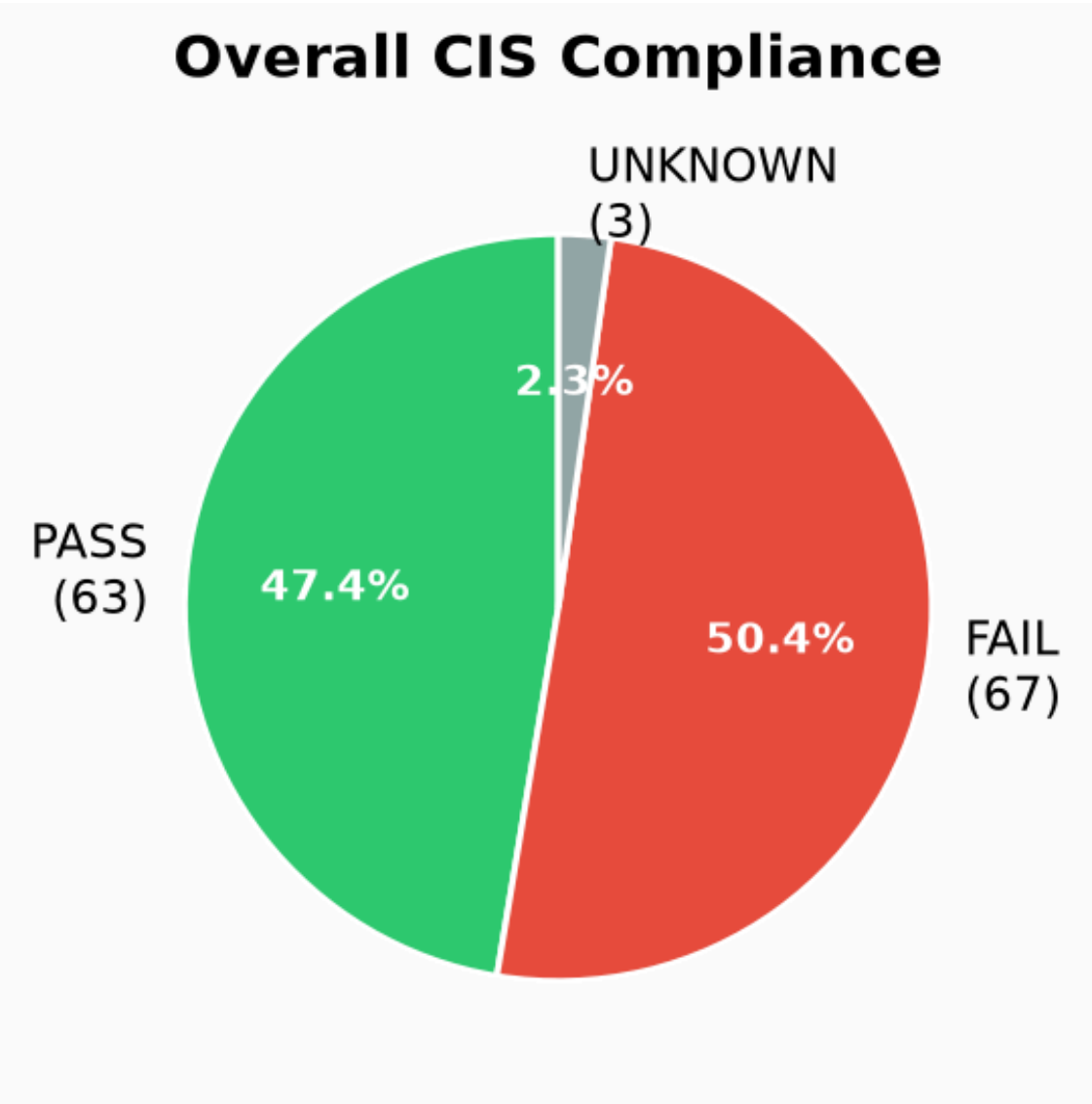


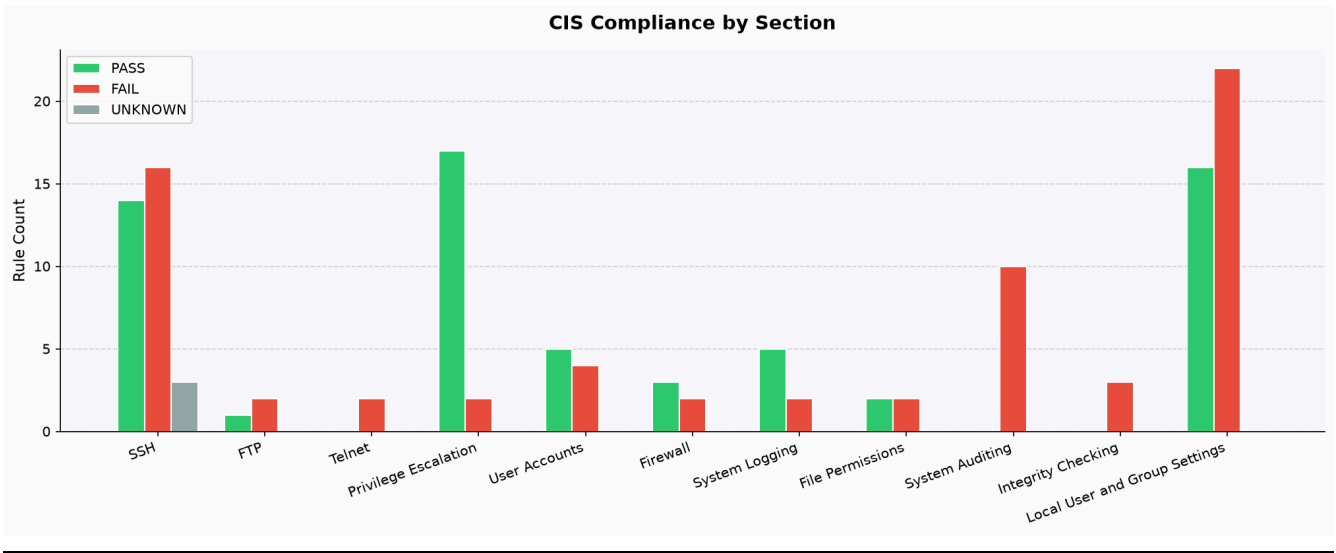
SENTINEL — CIS Benchmark Audit Report

Executive Summary

Metric	Value
Total Rules Evaluated	133
✓ Passed	63
✗ Failed	67
? Unknown	3
Overall Pass Rate	47.4%

Compliance Charts





Per-Section Breakdown

Section	✓ PASS	✗ FAIL	? UNKNOWN	Pass Rate
SSH	14	16	3	42%
FTP	1	2	0	33%
Telnet	0	2	0	0%
Privilege Escalation	17	2	0	89%
User Accounts	5	4	0	56%
Firewall	3	2	0	60%
System Logging	5	2	0	71%
File Permissions	2	2	0	50%
System Auditing	0	10	0	0%
Integrity Checking	0	3	0	0%
Local User and Group Settings	16	22	0	42%

CIS Benchmark Compliance Report

Here is the security compliance report for the remote machine based on the provided CIS Ubuntu 24.04 LTS Benchmark rules and security profile.

CIS Ubuntu 24.04 LTS Benchmark Compliance Report

SSH Rules

- Rule ID:** 1.6.5 **Rule Title:** Ensure sshd warning Banner is configured **Status:** FAIL
Evidence: `ssh.features.banner` is "none". **Recommendation:** Configure a warning banner by setting `Banner /etc/issue.net` in `/etc/ssh/sshd_config` and create the banner file.
- Rule ID:** 1.6.10 **Rule Title:** Ensure access to sshd warning banner is configured **Status:**

UNKNOWN Evidence: The Banner directive is not configured (Rule 1.6.5 FAIL), so there is no banner file to check permissions for. **Recommendation:** First, configure the SSH banner (see Rule 1.6.5). Then, ensure the banner file (e.g., /etc/issue.net) has permissions of 644 or more restrictive.

- **Rule ID:** 5.1.1 **Rule Title:** Ensure access to /etc/ssh/sshd_config is configured **Status:** FAIL **Evidence:** /etc/ssh/sshd_config has permissions 644 root root. **Recommendation:** Set permissions of /etc/ssh/sshd_config to 600 using sudo chmod 600 /etc/ssh/sshd_config.
- **Rule ID:** 5.1.2 **Rule Title:** Ensure access to SSH private host key files is configured **Status:** PASS **Evidence:** SSH private host key files (/etc/ssh/ssh_host_ecdsa_key, /etc/ssh/ssh_host_ed25519_key, /etc/ssh/ssh_host_rsa_key) all have permissions 600 root root. **Recommendation:** N/A
- **Rule ID:** 5.1.3 **Rule Title:** Ensure access to SSH public host key files is configured **Status:** PASS **Evidence:** SSH public host key files (/etc/ssh/ssh_host_ecdsa_key.pub, /etc/ssh/ssh_host_ed25519_key.pub, /etc/ssh/ssh_host_rsa_key.pub) all have permissions 644 root root. **Recommendation:** N/A
- **Rule ID:** 5.1.4 **Rule Title:** Ensure sshd access is configured **Status:** UNKNOWN **Evidence:** This is a general rule. Specific access controls are covered by other rules (e.g., PermitRootLogin, MaxAuthTries). **Recommendation:** N/A
- **Rule ID:** 5.1.5 **Rule Title:** Ensure sshd Banner is configured **Status:** FAIL **Evidence:** ssh.features.banner is "none". **Recommendation:** Configure a warning banner by setting Banner /etc/issue.net in /etc/ssh/sshd_config and create the banner file.
- **Rule ID:** 5.1.6 **Rule Title:** Ensure sshd Ciphers are configured **Status:** PASS **Evidence:** Configured ciphers are chacha20-poly1305@openssh.com, aes128-ctr, aes192-ctr, aes256-ctr, aes128-gcm@openssh.com, aes256-gcm@openssh.com, which are considered strong. **Recommendation:** N/A
- **Rule ID:** 5.1.7 **Rule Title:** Ensure sshd ClientAliveInterval and ClientAliveCountMax are configured **Status:** FAIL **Evidence:** ssh.session.client_alive_interval is 0, which disables client keepalives. **Recommendation:** Set ClientAliveInterval to a value between 300 and 900 seconds (e.g., ClientAliveInterval 300) and ClientAliveCountMax to 0-3 (e.g., ClientAliveCountMax 3) in /etc/ssh/sshd_config to terminate idle sessions.
- **Rule ID:** 5.1.8 **Rule Title:** Ensure sshd DisableForwarding is enabled **Status:** FAIL **Evidence:** ssh.forwarding.allow_tcp_forwarding is true, ssh.forwarding.allow_agent_forwarding is true, and ssh.features.x11_forwarding is true. **Recommendation:** Set AllowTcpForwarding no, AllowAgentForwarding no, and X11Forwarding no in /etc/ssh/sshd_config.
- **Rule ID:** 5.1.9 **Rule Title:** Ensure sshd GSSAPIAuthentication is disabled **Status:** PASS **Evidence:** ssh.authentication.gssapiauthentication is false. **Recommendation:** N/A
- **Rule ID:** 5.1.10 **Rule Title:** Ensure sshd HostbasedAuthentication is disabled **Status:**

PASS Evidence: `ssh.authentication.hostbasedauthentication` is false.

Recommendation: N/A

- **Rule ID:** 5.1.11 **Rule Title:** Ensure sshd IgnoreRhosts is enabled **Status:** PASS
Evidence: `ssh.authentication.ignorerhosts` is true. **Recommendation:** N/A
- **Rule ID:** 5.1.12 **Rule Title:** Ensure sshd KexAlgorithms is configured **Status:** FAIL
Evidence: The system has `crypto.weak_kex` algorithms enabled, including `ecdh-sha2-nistp256`, `ecdh-sha2-nistp384`, `ecdh-sha2-nistp521`, and `diffie-hellman-group14-sha256`. **Recommendation:** Explicitly configure `KexAlgorithms` in `/etc/ssh/sshd_config` to exclude weak algorithms. Refer to the CIS benchmark for a recommended list of strong KEX algorithms.
- **Rule ID:** 5.1.13 **Rule Title:** Ensure sshd LoginGraceTime is configured **Status:** FAIL
Evidence: `ssh.session.login_grace_time` is 120 seconds. **Recommendation:** Set `LoginGraceTime 60` or less in `/etc/ssh/sshd_config`.
- **Rule ID:** 5.1.14 **Rule Title:** Ensure sshd LogLevel is configured **Status:** PASS **Evidence:** `ssh.features.log_level` is "INFO". **Recommendation:** N/A
- **Rule ID:** 5.1.15 **Rule Title:** Ensure sshd MACs are configured **Status:** FAIL **Evidence:** The system has `crypto.weak_macs` algorithms enabled, including `umac-64-etm@openssh.com`, `hmac-sha1-etm@openssh.com` (Terrapin attack), `umac-64@openssh.com`, `umac-128@openssh.com`, `hmac-sha2-256`, `hmac-sha2-512`, and `hmac-sha1`. **Recommendation:** Explicitly configure MACs in `/etc/ssh/sshd_config` to exclude weak algorithms. Refer to the CIS benchmark for a recommended list of strong MAC algorithms.
- **Rule ID:** 5.1.16 **Rule Title:** Ensure sshd MaxAuthTries is configured **Status:** FAIL
Evidence: `ssh.authentication.max_auth_tries` is 6. **Recommendation:** Set `MaxAuthTries 4` or less in `/etc/ssh/sshd_config`.
- **Rule ID:** 5.1.17 **Rule Title:** Ensure sshd MaxStartups is configured **Status:** FAIL
Evidence: `ssh.session.maxstartups` is "10:30:100". The maximum allowed unauthenticated connections (100) is too high. **Recommendation:** Set `MaxStartups 10:30:60` or less in `/etc/ssh/sshd_config`.
- **Rule ID:** 5.1.18 **Rule Title:** Ensure sshd MaxSessions is configured **Status:** FAIL
Evidence: `ssh.session.max_sessions` is 10. **Recommendation:** Set `MaxSessions 4` or less in `/etc/ssh/sshd_config`.
- **Rule ID:** 5.1.19 **Rule Title:** Ensure sshd PermitEmptyPasswords is disabled **Status:** PASS **Evidence:** `ssh.authentication.permit_empty_passwords` is false.
Recommendation: N/A
- **Rule ID:** 5.1.20 **Rule Title:** Ensure sshd PermitRootLogin is disabled **Status:** FAIL
Evidence: `ssh.authentication.permit_root_login` is "without-password". **Recommendation:** Set `PermitRootLogin no` in `/etc/ssh/sshd_config` to prevent root login even with SSH keys.
- **Rule ID:** 5.1.21 **Rule Title:** Ensure sshd PermitUserEnvironment is disabled **Status:** PASS **Evidence:** `ssh.features.permit_user_environment` is false.
Recommendation: N/A
- **Rule ID:** 5.1.22 **Rule Title:** Ensure sshd UsePAM is enabled **Status:** PASS **Evidence:**

`ssh.authentication.usepam` is true. **Recommendation:** N/A

- **Rule ID:** 5.1.23 **Rule Title:** Ensure sshd post-quantum cryptography key exchange algorithms are configured **Status:** UNKNOWN **Evidence:** The provided security profile does not contain specific information about post-quantum cryptography key exchange algorithm configuration. **Recommendation:** Manually verify if post-quantum cryptography key exchange algorithms are configured according to the benchmark.
- **Rule ID:** 5.1.24 **Rule Title:** Ensure sshd ListenAddress is configured **Status:** FAIL **Evidence:** `ssh.network.listen_address` is "0.0.0.0:22", which listens on all interfaces. **Recommendation:** Set `ListenAddress <specific_IP_address>:22` in `/etc/ssh/sshd_config` to bind SSH to a specific network interface.

FTP Rules

- **Rule ID:** 2.1.8 **Rule Title:** Ensure ftp server services are not in use **Status:** FAIL **Evidence:** `ftp.service.installed` is "true", `ftp.service.enabled` is "true", and `ftp.service.running` is "true". The FTP server (vsftpd) is installed, enabled, and running. **Recommendation:** Uninstall the vsftpd package using `sudo apt purge vsftpd` if not required. If required, ensure it is securely configured (e.g., disable anonymous access, enable SSL/TLS, chroot users).
- **Rule ID:** 2.1.20 **Rule Title:** Ensure tftp server services are not in use **Status:** PASS **Evidence:** `ftp.tftp_server_packages` is "not installed" and `ftp.tftp_service_status` is "not running". **Recommendation:** N/A
- **Rule ID:** 2.2.6 **Rule Title:** Ensure ftp client is not installed **Status:** FAIL **Evidence:** The ftp client package is installed (`ftp client packages: ii ftp 20230507-2build3 all dummy transitional package for tnftp`). **Recommendation:** Uninstall the ftp client package using `sudo apt purge ftp`.

TELNET Rules

- **Rule ID:** 2.1.19 **Rule Title:** Ensure telnet server services are not in use **Status:** FAIL **Evidence:** `telnet.service.installed` is "true", `telnet.network.port_open` is "true", and `telnet.configuration.inetd_enabled` is "true". The telnet server is installed and configured to be active via inetd, and its port is open. **Recommendation:** Uninstall the telnet server package using `sudo apt purge telnetd` (or equivalent for the installed telnet server). Ensure port 23 is blocked by the firewall.
- **Rule ID:** 2.2.4 **Rule Title:** Ensure telnet client is not installed **Status:** FAIL **Evidence:** The telnet client package is installed (`telnet client packages: ii telnet 0.17+2.5-3ubuntu4.2 all transitional dummy package for inetutils-telnet default switch`). **Recommendation:** Uninstall the telnet client package using `sudo apt purge telnet`.

Privilege Escalation Rules

- **Rule ID:** 5.2.1 **Rule Title:** Ensure sudo is installed **Status:** PASS **Evidence:** `privilege_escalation.sudo_installed.installed` is true, with version "1.9.15p5". **Recommendation:** N/A
- **Rule ID:** 5.2.2 **Rule Title:** Ensure sudo commands use pty **Status:** PASS **Evidence:**

`privilege_escalation.use_pty_entries` contains "Defaults\tuse_pty" in `/etc/sudoers`. **Recommendation:** N/A

- **Rule ID:** 5.2.3 **Rule Title:** Ensure sudo log file exists **Status:** PASS **Evidence:** `privilege_escalation.sudo_syslog_logging.auth_log_exists` is true, `syslog_log_exists` is true, and `rsyslog_authpriv_configured` is true. Sudo logging is configured and active. **Recommendation:** N/A
- **Rule ID:** 5.2.4 **Rule Title:** Ensure users must provide password for escalation **Status:** PASS **Evidence:** `privilege_escalation.nopasswd_entries` is empty, indicating no NOPASSWD entries are configured. **Recommendation:** N/A
- **Rule ID:** 5.2.5 **Rule Title:** Ensure re-authentication for privilege escalation is not disabled globally **Status:** PASS **Evidence:** `privilege_escalation.noauthenticate_entries` is empty, indicating re-authentication is not disabled. **Recommendation:** N/A
- **Rule ID:** 5.2.6 **Rule Title:** Ensure sudo timestamp_timeout is configured **Status:** FAIL **Evidence:** `privilege_escalation.timestamp_timeout_entries` is empty, meaning `timestamp_timeout` is not explicitly configured and defaults to 15 minutes. **Recommendation:** Set `Defaults timestamp_timeout=0` or a short value (e.g., 5) in `/etc/sudoers` to require re-authentication for each sudo command or after a very short period.
- **Rule ID:** 5.2.7 **Rule Title:** Ensure access to the su command is restricted **Status:** FAIL **Evidence:** The `pam_wheel.so` line in `/etc/pam.d/su` is commented out (`# auth required pam_wheel.so`), meaning su access is not restricted to members of the wheel group. **Recommendation:** Uncomment `auth required pam_wheel.so` in `/etc/pam.d/su` and ensure only authorized users are members of the wheel group.

Directory Access Rules

- **Rule ID:** 7.1.1 **Rule Title:** Ensure access to `/etc/passwd` is configured **Status:** PASS **Evidence:** `/etc/passwd` has permissions `0644 root root`. **Recommendation:** N/A
- **Rule ID:** 7.1.2 **Rule Title:** Ensure access to `/etc/passwd-` is configured **Status:** PASS **Evidence:** `/etc/passwd-` has permissions `0644 root root`. **Recommendation:** N/A
- **Rule ID:** 7.1.3 **Rule Title:** Ensure access to `/etc/group` is configured **Status:** PASS **Evidence:** `/etc/group` has permissions `0644 root root`. **Recommendation:** N/A
- **Rule ID:** 7.1.4 **Rule Title:** Ensure access to `/etc/group-` is configured **Status:** PASS **Evidence:** `/etc/group-` has permissions `0644 root root`. **Recommendation:** N/A
- **Rule ID:** 7.1.5 **Rule Title:** Ensure access to `/etc/shadow` is configured **Status:** PASS **Evidence:** `/etc/shadow` has permissions `0640 root shadow`. **Recommendation:** N/A
- **Rule ID:** 7.1.6 **Rule Title:** Ensure access to `/etc/shadow-` is configured **Status:** PASS **Evidence:** `/etc/shadow-` has permissions `0640 root shadow`. **Recommendation:** N/A
- **Rule ID:** 7.1.7 **Rule Title:** Ensure access to `/etc/gshadow` is configured **Status:** PASS

Evidence: /etc/gshadow has permissions 0640 root shadow. **Recommendation:** N/A

- **Rule ID:** 7.1.8 **Rule Title:** Ensure access to /etc/gshadow- is configured **Status:** PASS
Evidence: /etc/gshadow- has permissions 0640 root shadow.
Recommendation: N/A
- **Rule ID:** 7.1.9 **Rule Title:** Ensure access to /etc/shells is configured **Status:** PASS
Evidence: /etc/shells has permissions 0644 root root. **Recommendation:** N/A
- **Rule ID:** 7.1.10 **Rule Title:** Ensure access to /etc/security/opasswd is configured
Status: PASS **Evidence:** /etc/security/opasswd has permissions 0600 root root. **Recommendation:** N/A
- **Rule ID:** 7.1.11 **Rule Title:** Ensure world writable files and directories are secured
Status: PASS **Evidence:**
file_permissions.world_writable.world_writable_files and
file_permissions.world_writable.world_writable_dirs_no_sticky are
both empty for scanned mounts. **Recommendation:** N/A
- **Rule ID:** 7.1.12 **Rule Title:** Ensure no files or directories without an owner and a group
exist **Status:** PASS **Evidence:**
file_permissions.unowned.unowned_or_ungrouped is empty for scanned
mounts. **Recommendation:** N/A
- **Rule ID:** 7.1.13 **Rule Title:** Ensure SUID and SGID files are reviewed **Status:**
INFORMATIONAL **Evidence:** The system lists several SUID/SGID files, including /
usr/bin/umount, /usr/bin/chsh, /usr/bin/sudo, /usr/bin/su, /usr/bin/
passwd, /usr/bin/ssh-agent, /usr/bin/fusermount3, /usr/bin/newgrp, /
usr/bin/chage, /usr/bin/expiry, /usr/bin/gpasswd, /usr/bin/crontab, /
usr/bin/mount, /usr/bin/chfn, /usr/lib/dbus-1.0/dbus-daemon-launch-
helper, /usr/lib/x86_64-linux-gnu/utempter/utempter, /usr/lib/
openssh/ssh-keysign, /usr/lib/polkit-1/polkit-agent-helper-1, /usr/
sbin/pam_extrausers_chkpwd, /usr/sbin/unix_chkpwd. **Recommendation:**
Regularly review all SUID/SGID files to ensure they are legitimate and necessary for
system operation. Remove or restrict any unnecessary SUID/SGID binaries.

User Accounts Rules

- **Rule ID:** 5.4.1.1 **Rule Title:** Ensure password expiration is configured **Status:** FAIL
Evidence:
user_accounts.shadow_password_suite.login_defs.PASS_MAX_DAYS is
"99999", meaning passwords never expire. **Recommendation:** Set PASS_MAX_DAYS to
365 or less in /etc/login.defs to enforce regular password changes.
- **Rule ID:** 5.4.1.2 **Rule Title:** Ensure minimum password days is configured **Status:**
INFORMATIONAL **Evidence:**
user_accounts.shadow_password_suite.login_defs.PASS_MIN_DAYS is "0",
allowing immediate password changes. **Recommendation:** Consider setting
PASS_MIN_DAYS to 1 or more in /etc/login.defs to prevent users from cycling
through passwords to reuse a previous one.
- **Rule ID:** 5.4.1.3 **Rule Title:** Ensure password expiration warning days is configured
Status: PASS **Evidence:**

`user_accounts.shadow_password_suite.login_defs.PASS_WARN_AGE` is "7".

Recommendation: N/A

- **Rule ID:** 5.4.1.4 **Rule Title:** Ensure strong password hashing algorithm is configured **Status:** PASS **Evidence:** `user_accounts.shadow_password_suite.login_defs.ENCRYPT_METHOD` is "SHA512" and `pam_unix_hashing` shows "yescrypt". Both are strong hashing algorithms. **Recommendation:** N/A
- **Rule ID:** 5.4.1.5 **Rule Title:** Ensure inactive password lock is configured **Status:** FAIL **Evidence:** `user_accounts.shadow_password_suite.useradd_inactive` is "-1" and `user_chage_info` shows "account_inactive": "never" for all users. **Recommendation:** Set `INACTIVE` to 30 or less in `/etc/default/useradd` and apply to existing users with `chage -I <days> <username>` to lock accounts after a period of inactivity.
- **Rule ID:** 5.4.1.6 **Rule Title:** Ensure all users last password change date is in the past **Status:** FAIL **Evidence:** Several users (e.g., root, daemon, bin, sys, sync, games, man, lp, mail, news, uucp, proxy, www-data, backup, list, irc, _apt, nobody, systemd-network, systemd-timesync, dhcpcd, messagebus, systemd-resolve, pollinate, polkitd, syslog, uidd, tcpdump, tss, landscape, fwupd-refresh) have `last_password_change` dates in the future (e.g., "Feb 16, 2025", "Apr 19, 2026", "Jun 25, 2026"). **Recommendation:** Force password changes for all users with future last password change dates using `sudo chage -d 0 <username>`.
- **Rule ID:** 5.4.2.1 **Rule Title:** Ensure root is the only UID 0 account **Status:** PASS **Evidence:** `user_accounts.root_and_system_accounts.uid_0_accounts` lists only "root". **Recommendation:** N/A
- **Rule ID:** 5.4.2.2 **Rule Title:** Ensure root is the only GID 0 account **Status:** PASS **Evidence:** `user_accounts.root_and_system_accounts.primary_gid_0_accounts` lists only "root". **Recommendation:** N/A
- **Rule ID:** 5.4.2.3 **Rule Title:** Ensure group root is the only GID 0 group **Status:** PASS **Evidence:** `user_accounts.root_and_system_accounts.gid_0_groups` lists only "root". **Recommendation:** N/A
- **Rule ID:** 5.4.2.4 **Rule Title:** Ensure root account access is controlled **Status:** FAIL **Evidence:** `ssh.authentication.permit_root_login` is "without-password" (allowing key-based root login via SSH) and `user_accounts.root_and_system_accounts.securetty_exists` is false (allowing root login on any TTY if direct login is enabled). **Recommendation:** Set `PermitRootLogin no` in `/etc/ssh/sshd_config`. If direct console root login is necessary, create and configure `/etc/securetty` to restrict root logins to specific TTYS.
- **Rule ID:** 5.4.2.5 **Rule Title:** Ensure root path integrity **Status:** FAIL **Evidence:** `user_accounts.root_and_system_accounts.root_path` includes non-standard and potentially insecure directories like `/usr/games`, `/usr/local/games`, and `/snap/bin`. **Recommendation:** Review and restrict root's `PATH` environment variable to essential and secure directories (e.g., `/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin`).

- **Rule ID:** 5.4.2.6 **Rule Title:** Ensure root user umask is configured **Status:** PASS
Evidence: `user_accounts.root_and_system_accounts.root_umask.login_defs` is "022". **Recommendation:** N/A
- **Rule ID:** 5.4.2.7 **Rule Title:** Ensure system accounts do not have a valid login shell **Status:** FAIL **Evidence:** The `sync` system account has `/bin/sync` as its login shell, which is a valid shell. **Recommendation:** Change the login shell for the `sync` user to `/usr/sbin/nologin` or `/bin/false` using `sudo usermod -s /usr/sbin/nologin sync`.
- **Rule ID:** 5.4.2.8 **Rule Title:** Ensure accounts without a valid login shell are locked **Status:** PASS **Evidence:** All listed system accounts in `user_accounts.root_and_system_accounts.system_accounts` have `passwd_status` as "L" (locked). **Recommendation:** N/A
- **Rule ID:** 5.4.3.1 **Rule Title:** Ensure `nologin` is not listed in `/etc/shells` **Status:** PASS
Evidence: `user_accounts.default_user_environment.nologin_in_shells` is false. **Recommendation:** N/A
- **Rule ID:** 5.4.3.2 **Rule Title:** Ensure default user shell timeout is configured **Status:** FAIL **Evidence:** `user_accounts.default_user_environment.tmount_settings` shows no `TMOUT` configuration in `/etc/profile`, `/etc/profile.d`, or `/etc/bash.bashrc`. **Recommendation:** Set `TMOUT` to 600 seconds or less in a global shell configuration file (e.g., `/etc/profile.d/timeout.sh`) to automatically log out idle users.
- **Rule ID:** 5.4.3.3 **Rule Title:** Ensure default user umask is configured **Status:** PASS
Evidence: `user_accounts.default_user_environment.default_user_umask.login_def` is "022" and `pam_umask` module is enabled. **Recommendation:** N/A

Firewall Rules

- **Rule ID:** 4.1.1 **Rule Title:** Ensure `ufw` is installed **Status:** PASS **Evidence:** `ufw_firewall.ufw_installed` is true. **Recommendation:** N/A
- **Rule ID:** 4.1.2 **Rule Title:** Ensure `ufw` service is configured **Status:** FAIL **Evidence:** `ufw_firewall.service_configured.is_enabled` is "enabled" and `is_active` is "active", but `ufw_firewall.service_configured.status_verbose` reports "Status: inactive". This indicates UFW is not actively running despite being enabled. **Recommendation:** Ensure UFW is enabled and active by running `sudo ufw enable` and verifying its status.
- **Rule ID:** 4.1.3 **Rule Title:** Ensure `ufw` incoming default is configured **Status:** PASS
Evidence: `ufw_firewall.default_policies.etc_default_ufw.DEFAULT_INPUT_POLICY` is "DROP". **Recommendation:** N/A
- **Rule ID:** 4.1.4 **Rule Title:** Ensure `ufw` outgoing default is configured **Status:** FAIL
Evidence: `ufw_firewall.default_policies.etc_default_ufw.DEFAULT_OUTPUT_POLICY` is "ACCEPT". **Recommendation:** Set `DEFAULT_OUTPUT_POLICY="DROP"` in `/etc/default/ufw` and configure specific allow rules for necessary outgoing traffic.

- **Rule ID:** 4.1.5 **Rule Title:** Ensure ufw routed default is configured **Status:** PASS
Evidence: `ufw_firewall.default_policies.etc_default_ufw.DEFAULT_FORWARD_POLICY` is "DROP". **Recommendation:** N/A

System Logging Rules

- **Rule ID:** 6.1.1.1.1 **Rule Title:** Ensure systemd-journald service is enabled and active **Status:** PASS **Evidence:** `system_logging.journald.systemd-journald_active` is "active" and `systemd-journald_enabled` is "static". **Recommendation:** N/A
- **Rule ID:** 6.1.1.1.2 **Rule Title:** Ensure systemd-journal-remote service is not in use **Status:** PASS **Evidence:** `system_logging.journald.systemd-journal-remote_installed` is false. **Recommendation:** N/A
- **Rule ID:** 6.1.1.1.3 **Rule Title:** Ensure journald is configured to send logs to rsyslog **Status:** FAIL **Evidence:** The `ForwardToSyslog` option in `/etc/systemd/journald.conf` is commented out (`#ForwardToSyslog=no`), meaning it defaults to no. **Recommendation:** Uncomment and set `ForwardToSyslog=yes` in `/etc/systemd/journald.conf` and restart `systemd-journald`.
- **Rule ID:** 6.1.1.1.4 **Rule Title:** Ensure journald log file access is configured **Status:** INFORMATIONAL **Evidence:** Journald log directories `/run/log/journal` and `/var/log/journal` have permissions `2755 root 999`. **Recommendation:** N/A (Manual review required to confirm compliance with specific access requirements).
- **Rule ID:** 6.1.1.1.5 **Rule Title:** Ensure journald log file rotation is configured **Status:** INFORMATIONAL **Evidence:** Log rotation settings like `SystemMaxUse`, `SystemKeepFree`, `SystemMaxFileSize`, `SystemMaxFiles`, `MaxRetentionSec`, `MaxFileSec` are all commented out in `/etc/systemd/journald.conf`, relying on default values. **Recommendation:** N/A (Manual review required to confirm default rotation settings are adequate for compliance).
- **Rule ID:** 6.1.1.1.6 **Rule Title:** Ensure journald Storage is configured **Status:** FAIL **Evidence:** The `Storage` option in `/etc/systemd/journald.conf` is commented out (`#Storage=auto`), meaning it defaults to `auto`. This may result in volatile storage if `/var/log/journal` is not present. **Recommendation:** Uncomment and set `Storage=persistent` in `/etc/systemd/journald.conf` to ensure logs are stored persistently.
- **Rule ID:** 6.1.1.1.7 **Rule Title:** Ensure journald Compress is configured **Status:** PASS **Evidence:** The `Compress` option in `/etc/systemd/journald.conf` is commented out (`#Compress=yes`), meaning it defaults to `yes`. **Recommendation:** N/A
- **Rule ID:** 6.1.2.1 **Rule Title:** Ensure rsyslog is installed **Status:** PASS **Evidence:** `system_logging.rsyslog.rsyslog_installed` is true. **Recommendation:** N/A
- **Rule ID:** 6.1.2.2 **Rule Title:** Ensure rsyslog service is enabled and active **Status:** PASS **Evidence:** `system_logging.rsyslog.rsyslog_active` is "active" and `rsyslog_enabled` is "enabled". **Recommendation:** N/A
- **Rule ID:** 6.1.2.3 **Rule Title:** Ensure rsyslog default file permissions are configured **Status:** PASS **Evidence:** `/etc/rsyslog.conf` sets `$FileOwner syslog`,

\$FileGroup adm,\$FileCreateMode 0640,\$DirCreateMode 0755,\$Umask 0022. **Recommendation:** N/A

- **Rule ID:** 6.1.2.4 **Rule Title:** Ensure rsyslog logging is configured **Status:** INFORMATIONAL **Evidence:** Rsyslog configuration files (/etc/rsyslog.conf, /etc/rsyslog.d/*.conf) are present and contain various logging rules. **Recommendation:** N/A (Manual review required to confirm all necessary logging is configured).
- **Rule ID:** 6.1.2.5 **Rule Title:** Ensure rsyslog is configured to send logs to a remote log host **Status:** INFORMATIONAL **Evidence:** The provided rsyslog configuration does not show explicit rules for sending logs to a remote log host. **Recommendation:** N/A (Manual review required to confirm if remote logging is intended and configured).
- **Rule ID:** 6.1.2.6 **Rule Title:** Ensure rsyslog is not configured to receive logs from a remote client **Status:** PASS **Evidence:** module(load="imudp") and module(load="imtcp") are commented out in /etc/rsyslog.conf. **Recommendation:** N/A
- **Rule ID:** 6.1.2.7 **Rule Title:** Ensure logrotate is configured **Status:** INFORMATIONAL **Evidence:** logrotate is installed, logrotate_timer_active is "active", logrotate_timer_enabled is "enabled", and various logrotate configuration files are present. **Recommendation:** N/A (Manual review required to confirm log rotation settings are adequate for all critical logs).
- **Rule ID:** 6.1.2.8 **Rule Title:** Ensure rsyslog-gnutls is installed **Status:** FAIL **Evidence:** system_logging.rsyslog.rsyslog_gnutls_installed is false. **Recommendation:** Install the rsyslog-gnutls package using `sudo apt install rsyslog-gnutls`.
- **Rule ID:** 6.1.2.9 **Rule Title:** Ensure rsyslog is configured to use gtls for forwarding **Status:** FAIL **Evidence:** rsyslog-gnutls is not installed, so rsyslog cannot be configured to use GTLS for forwarding. **Recommendation:** Install rsyslog-gnutls and configure rsyslog to use GTLS for secure remote log forwarding.
- **Rule ID:** 6.1.2.10 **Rule Title:** Ensure rsyslog CA certificates are configured **Status:** INFORMATIONAL **Evidence:** rsyslog-gnutls is not installed, so CA certificates for GTLS are not relevant. **Recommendation:** N/A (Manual review required if GTLS is intended for remote logging).
- **Rule ID:** 6.1.3.1 **Rule Title:** Ensure access to all logfiles has been configured **Status:** INFORMATIONAL **Evidence:** system_logging.logfiles lists various log files with their permissions. **Recommendation:** N/A (Manual review required to confirm all log files have appropriate permissions).

System Auditing Rules

- **Rule ID:** 6.2.1.1.1 **Rule Title:** Ensure auditd is installed **Status:** FAIL **Evidence:** system_auditing.auditd_installed is false. **Recommendation:** Install the auditd package using `sudo apt install auditd`.
- **Rule ID:** 6.2.1.1.2 **Rule Title:** Ensure auditd service is enabled and active **Status:** FAIL **Evidence:** auditd is not installed, so the service cannot be enabled or active. **Recommendation:** Install auditd and then enable and start the auditd service using

`sudo systemctl enable auditd` and `sudo systemctl start auditd`.

- **Rule ID:** 6.2.1.1.3 **Rule Title:** Ensure auditing for processes that start prior to auditd is enabled **Status:** FAIL **Evidence:** auditd is not installed. **Recommendation:** Install auditd and configure GRUB to include `audit=1` in the kernel command line.
- **Rule ID:** 6.2.1.1.4 **Rule Title:** Ensure `audit_backlog_limit` is sufficient **Status:** FAIL **Evidence:** auditd is not installed. **Recommendation:** Install auditd and set `audit_backlog_limit` to 8192 or greater in `/etc/audit/audit.conf`.
- **Rule ID:** 6.2.1.2.1 **Rule Title:** Ensure audit log storage size is configured **Status:** FAIL **Evidence:** auditd is not installed. **Recommendation:** Install auditd and configure `max_log_file` in `/etc/audit/auditd.conf` to an appropriate size (e.g., 50MB).
- **Rule ID:** 6.2.1.2.2 **Rule Title:** Ensure audit logs are not automatically deleted **Status:** FAIL **Evidence:** auditd is not installed. **Recommendation:** Install auditd and set `max_log_file_action = keep_logs` in `/etc/audit/auditd.conf`.
- **Rule ID:** 6.2.1.2.3 **Rule Title:** Ensure system is disabled when audit logs are full **Status:** FAIL **Evidence:** auditd is not installed. **Recommendation:** Install auditd and set `space_left_action = SYSLOG`, `action_mail_acct = root`, and `admin_space_left_action = HALT` in `/etc/audit/auditd.conf`.
- **Rule ID:** 6.2.1.2.4 **Rule Title:** Ensure system warns when audit logs are low on space **Status:** FAIL **Evidence:** auditd is not installed. **Recommendation:** Install auditd and set `space_left_action = SYSLOG` and `action_mail_acct = root` in `/etc/audit/auditd.conf`.
- **Rule ID:** 6.2.3.1-6.2.3.30 **Rule Title:** Ensure auditd rules cover sudoers, identity, logins, session, time-change, MAC policy, access, mounts, deletion, loading modules, and configuration is immutable **Status:** FAIL **Evidence:** auditd is not installed, so no audit rules are configured or enforced. **Recommendation:** Install auditd and configure the comprehensive set of audit rules as specified in the CIS benchmark in `/etc/audit/rules.d/`.
- **Rule ID:** 6.2.4.1-6.2.4.10 **Rule Title:** Ensure audit log files and audit tools file access and permissions are strictly constrained **Status:** FAIL **Evidence:** auditd is not installed, so audit log files and tools are not present or managed by auditd. **Recommendation:** Install auditd and ensure that audit log files (e.g., `/var/log/audit/audit.log`) and audit tools (e.g., `/sbin/auditctl`) have appropriate restrictive permissions (e.g., 0640 for logs, 0755 for tools).

Integrity Checking Rules

- **Rule ID:** 6.3.1 **Rule Title:** Ensure AIDE is installed **Status:** FAIL **Evidence:** `aide_integrity_checking.aide_installed` is false. **Recommendation:** Install the aide package using `sudo apt install aide`.
- **Rule ID:** 6.3.2 **Rule Title:** Ensure filesystem integrity is regularly checked **Status:** FAIL **Evidence:** aide is not installed, so filesystem integrity checks are not configured. **Recommendation:** Install aide and configure a cron job or systemd timer to regularly run `aide --check`.
- **Rule ID:** 6.3.3 **Rule Title:** Ensure cryptographic mechanisms protect tools **Status:** FAIL

Evidence: aide is not installed, so cryptographic integrity checking for tools is not configured. **Recommendation:** Install aide and configure aide.conf to include critical system binaries with strong hashing algorithms (e.g., p+i+n+u+g+s+b+acl+xattrs+sha512).

Local User and Group Settings Rules

- **Rule ID:** 7.2.1 **Rule Title:** Ensure accounts in /etc/passwd use shadowed passwords **Status:** PASS **Evidence:** user_accounts.shadow_password_suite.pam_unix_hashing shows pam_unix.so obscure yescrypt, indicating shadowed passwords are in use. **Recommendation:** N/A
- **Rule ID:** 7.2.2 **Rule Title:** Ensure /etc/shadow password fields are not empty **Status:** PASS **Evidence:** The user_chage_info entries show last password change dates, implying password fields are not empty. **Recommendation:** N/A
- **Rule ID:** 7.2.3 **Rule Title:** Ensure all groups in /etc/passwd exist in /etc/group **Status:** PASS **Evidence:** user_accounts.local_users_and_groups_evidence.orphan_gids_in_passwd is empty. **Recommendation:** N/A
- **Rule ID:** 7.2.4 **Rule Title:** Ensure shadow group is empty **Status:** PASS **Evidence:** user_accounts.local_users_and_groups_evidence.shadow_group_members is empty. **Recommendation:** N/A
- **Rule ID:** 7.2.5 **Rule Title:** Ensure no duplicate UIDs exist **Status:** PASS **Evidence:** user_accounts.local_users_and_groups_evidence.duplicate_uids is empty. **Recommendation:** N/A
- **Rule ID:** 7.2.6 **Rule Title:** Ensure no duplicate GIDs exist **Status:** PASS **Evidence:** user_accounts.local_users_and_groups_evidence.duplicate_gids is empty. **Recommendation:** N/A
- **Rule ID:** 7.2.7 **Rule Title:** Ensure no duplicate user names exist **Status:** PASS **Evidence:** user_accounts.local_users_and_groups_evidence.duplicate_usernames is empty. **Recommendation:** N/A
- **Rule ID:** 7.2.8 **Rule Title:** Ensure no duplicate group names exist **Status:** PASS **Evidence:** user_accounts.local_users_and_groups_evidence.duplicate_groupnames is empty. **Recommendation:** N/A
- **Rule ID:** 7.2.9 **Rule Title:** Ensure local interactive user home directories are configured **Status:** PASS **Evidence:** user_accounts.local_users_and_groups_evidence.home_directory_issues is empty. **Recommendation:** N/A
- **Rule ID:** 7.2.10 **Rule Title:** Ensure local interactive user dot files access is configured **Status:** PASS **Evidence:** user_accounts.local_users_and_groups_evidence.dotfile_permission_is is empty. **Recommendation:** N/A

SECTION 3 – NETWORK CONFIGURATION

- **Rule ID:** 3.1.1 **Rule Title:** IPv6 status identified **Status:** INFORMATIONAL **Evidence:** sysfs_disable_value: "0" (IPv6 enabled), grub_ipv6_disable_flag: false (not disabled in GRUB), inet6_address_count: 2 (IPv6 active on 2 interfaces). **Recommendation:** N/A
- **Rule ID:** 3.1.2 **Rule Title:** Ensure wireless interfaces are not available **Status:** FAIL **Evidence:** network_config.network_devices.wireless_interfaces.loaded_wireless_contains "cfg80211", and network_config.network_devices.wireless_interfaces.any_wireless_found is true. **Recommendation:** Disable or remove wireless interfaces and blacklist wireless kernel modules (e.g., install cfg80211 /bin/false, blacklist cfg80211 in /etc/modprobe.d/CIS.conf).
- **Rule ID:** 3.1.3 **Rule Title:** Ensure Bluetooth services are not in use **Status:** PASS **Evidence:** network_config.network_devices.bluetooth.bluez_installed is false. **Recommendation:** N/A
- **Rule ID:** 3.2 Network Kernel Modules **Rule Title:** Ensure unused network protocols are disabled **Status:** FAIL (for all listed modules) **Evidence:**
 - **atm:** currently_loaded: false, modprobe_config_lines: []. Missing disable/blacklist.
 - **can:** currently_loaded: false, modprobe_config_lines: []. Missing disable/blacklist.
 - **dccp:** currently_loaded: false, modprobe_config_lines: []. Missing disable/blacklist.
 - **rds:** currently_loaded: false, modprobe_config_lines: []. Missing disable/blacklist.
 - **sctp:** currently_loaded: false, modprobe_config_lines: []. Missing disable/blacklist.
 - **tipc:** currently_loaded: false, modprobe_config_lines: []. Missing disable/blacklist. **Recommendation:** For each module (atm, can, dccp, rds, sctp, tipc), add install <module> /bin/false and blacklist <module> to a modprobe configuration file (e.g., /etc/modprobe.d/CIS.conf).
- **Rule ID:** 3.3.1 IPv4 Kernel Parameters **Rule Title:** Ensure IPv4 kernel parameters are configured **Status:** Mixed (see individual parameters below) **Evidence & Recommendation:**
 - net.ipv4.ip_forward: runtime_value: "0", persisted_value: null. **FAIL.** **Recommendation:** Add net.ipv4.ip_forward = 0 to a sysctl configuration file (e.g., /etc/sysctl.d/99-cis-sysctl.conf).
 - net.ipv4.conf.all.forwarding: runtime_value: "0", persisted_value: null. **FAIL.** **Recommendation:** Add net.ipv4.conf.all.forwarding = 0 to a sysctl configuration file.
 - net.ipv4.conf.default.forwarding: runtime_value: "0", persisted_value: null. **FAIL.** **Recommendation:** Add net.ipv4.conf.default.forwarding = 0 to a sysctl configuration file.
 - net.ipv4.conf.all.send_redirects: runtime_value: "1", persisted_value: null. **FAIL.** **Recommendation:** Set net.ipv4.conf.all.send_redirects = 0 and persist it in a sysctl

configuration file.

- `net.ipv4.conf.default.send_redirects`: runtime_value: "1", persisted_value: null. **FAIL. Recommendation: Set**
`net.ipv4.conf.default.send_redirects = 0` and persist it in a sysctl configuration file.
- `net.ipv4.icmp_ignore_bogus_error_responses`: runtime_value: "1", persisted_value: null. **FAIL. Recommendation: Add**
`net.ipv4.icmp_ignore_bogus_error_responses = 1` to a sysctl configuration file.
- `net.ipv4.icmp_echo_ignore_broadcasts`: runtime_value: "1", persisted_value: null. **FAIL. Recommendation: Add**
`net.ipv4.icmp_echo_ignore_broadcasts = 1` to a sysctl configuration file.
- `net.ipv4.conf.all.accept_redirects`: runtime_value: "1", persisted_value: null. **FAIL. Recommendation: Set**
`net.ipv4.conf.all.accept_redirects = 0` and persist it in a sysctl configuration file.
- `net.ipv4.conf.default.accept_redirects`: runtime_value: "1", persisted_value: null. **FAIL. Recommendation: Set**
`net.ipv4.conf.default.accept_redirects = 0` and persist it in a sysctl configuration file.
- `net.ipv4.conf.all.secure_redirects`: runtime_value: "1", persisted_value: null. **FAIL. Recommendation: Set**
`net.ipv4.conf.all.secure_redirects = 0` and persist it in a sysctl configuration file.
- `net.ipv4.conf.default.secure_redirects`: runtime_value: "1", persisted_value: null. **FAIL. Recommendation: Set**
`net.ipv4.conf.default.secure_redirects = 0` and persist it in a sysctl configuration file.
- `net.ipv4.conf.all.rp_filter`: runtime_value: "2", persisted_value: "2". **PASS. Recommendation: N/A.**
- `net.ipv4.conf.default.rp_filter`: runtime_value: "2", persisted_value: "2". **PASS. Recommendation: N/A.**
- `net.ipv4.conf.all.accept_source_route`: runtime_value: "0", persisted_value: null. **FAIL. Recommendation: Add**
`net.ipv4.conf.all.accept_source_route = 0` to a sysctl configuration file.
- `net.ipv4.conf.default.accept_source_route`: runtime_value: "1", persisted_value: null. **FAIL. Recommendation: Set**
`net.ipv4.conf.default.accept_source_route = 0` and persist it in a sysctl configuration file.
- `net.ipv4.conf.all.log_martians`: runtime_value: "0", persisted_value: null. **FAIL. Recommendation: Set**
`net.ipv4.conf.all.log_martians = 1` and persist it in a sysctl configuration file.
- `net.ipv4.conf.default.log_martians`: runtime_value: "0", persisted_value: null. **FAIL. Recommendation: Set**
`net.ipv4.conf.default.log_martians = 1` and persist it in a sysctl configuration file.
- `net.ipv4.tcp_syncookies`: runtime_value: "1", persisted_value: null. **FAIL. Recommendation: Add**
`net.ipv4.tcp_syncookies = 1` to a sysctl configuration file.

- **Rule ID:** 3.3.2 IPv6 Kernel Parameters **Rule Title:** Ensure IPv6 kernel parameters are configured **Status:** Mixed (see individual parameters below) **Evidence & Recommendation:**

- `net.ipv6.conf.all.forwarding`: runtime_value: "0", persisted_value: null. **FAIL. Recommendation:** Add `net.ipv6.conf.all.forwarding = 0` to a sysctl configuration file.
- `net.ipv6.conf.default.forwarding`: runtime_value: "0", persisted_value: null. **FAIL. Recommendation:** Add `net.ipv6.conf.default.forwarding = 0` to a sysctl configuration file.
- `net.ipv6.conf.all.accept_redirects`: runtime_value: "1", persisted_value: null. **FAIL. Recommendation:** Set `net.ipv6.conf.all.accept_redirects = 0` and persist it in a sysctl configuration file.
- `net.ipv6.conf.default.accept_redirects`: runtime_value: "1", persisted_value: null. **FAIL. Recommendation:** Set `net.ipv6.conf.default.accept_redirects = 0` and persist it in a sysctl configuration file.
- `net.ipv6.conf.all.accept_source_route`: runtime_value: "0", persisted_value: null. **FAIL. Recommendation:** Add `net.ipv6.conf.all.accept_source_route = 0` to a sysctl configuration file.
- `net.ipv6.conf.default.accept_source_route`: runtime_value: "0", persisted_value: null. **FAIL. Recommendation:** Add `net.ipv6.conf.default.accept_source_route = 0` to a sysctl configuration file.
- `net.ipv6.conf.all.accept_ra`: runtime_value: "1", persisted_value: null. **FAIL. Recommendation:** Set `net.ipv6.conf.all.accept_ra = 0` and persist it in a sysctl configuration file.
- `net.ipv6.conf.default.accept_ra`: runtime_value: "1", persisted_value: null. **FAIL. Recommendation:** Set `net.ipv6.conf.default.accept_ra = 0` and persist it in a sysctl configuration file.

SECTION 5.3 – PLUGGABLE AUTHENTICATION MODULES (PAM)

- **Rule ID:** 5.3.1.1 **Rule Title:** libpam-runtime: Ensure PAM package is up-to-date **Status:** PASS **Evidence:** libpam-runtime is installed (version 1.5.3-5ubuntu5.5) and matches the candidate version. **Recommendation:** N/A
- **Rule ID:** 5.3.1.2 **Rule Title:** libpam-modules: Ensure PAM package is up-to-date **Status:** PASS **Evidence:** libpam-modules is installed (version 1.5.3-5ubuntu5.5) and matches the candidate version. **Recommendation:** N/A
- **Rule ID:** 5.3.1.3 **Rule Title:** libpam-pwquality: Ensure PAM package is installed **Status:** FAIL **Evidence:** libpam-pwquality is not installed. **Recommendation:** Install the libpam-pwquality package using `sudo apt install libpam-pwquality`.
- **Rule ID:** 5.3.1.4 **Rule Title:** cracklib-runtime: Ensure PAM package is installed **Status:** FAIL **Evidence:** cracklib-runtime is not installed. **Recommendation:** Install the cracklib-runtime package using `sudo apt install cracklib-runtime`.

- **Rule ID:** 5.3.2.1 **Rule Title:** pam_unix enabled **Status:** PASS **Evidence:** pam.pam_profiles_raw.module_lines.pam_unix.so contains active lines in common-auth, common-account, common-password, and common-session. **Recommendation:** N/A
- **Rule ID:** 5.3.2.2 **Rule Title:** pam_faillock enabled **Status:** FAIL **Evidence:** pam.pam_profiles_raw.module_lines.pam_faillock.so is empty, indicating pam_faillock is not enabled. **Recommendation:** Enable pam_faillock.so in PAM configuration using sudo pam-auth-update or by manually adding lines to relevant /etc/pam.d/ files.
- **Rule ID:** 5.3.2.3 **Rule Title:** pam_pwquality enabled **Status:** FAIL **Evidence:** pam.pam_profiles_raw.module_lines.pam_pwquality.so is empty, indicating pam_pwquality is not enabled. **Recommendation:** Enable pam_pwquality.so in PAM configuration using sudo pam-auth-update or by manually adding lines to relevant /etc/pam.d/ files.
- **Rule ID:** 5.3.2.4 **Rule Title:** pam_pwhistory enabled **Status:** FAIL **Evidence:** pam.pam_profiles_raw.module_lines.pam_pwhistory.so is empty, indicating pam_pwhistory is not enabled. **Recommendation:** Enable pam_pwhistory.so in PAM configuration using sudo pam-auth-update or by manually adding lines to relevant /etc/pam.d/ files.
- **Rule ID:** 5.3.3.1.1 **Rule Title:** pam_faillock arguments: deny <= 5 **Status:** FAIL **Evidence:** pam.pam_faillock.faillock_conf_settings.deny is null, and pam_faillock is not enabled. **Recommendation:** Enable pam_faillock (see 5.3.2.2) and set deny = 5 or less in /etc/security/faillock.conf.
- **Rule ID:** 5.3.3.1.2 **Rule Title:** pam_faillock arguments: unlock_time >= 900 **Status:** FAIL **Evidence:** pam.pam_faillock.faillock_conf_settings.unlock_time is null, and pam_faillock is not enabled. **Recommendation:** Enable pam_faillock (see 5.3.2.2) and set unlock_time = 900 or 0 (never unlock) in /etc/security/faillock.conf.
- **Rule ID:** 5.3.3.1.3 **Rule Title:** pam_faillock arguments: even_deny_root / root_unlock_time **Status:** FAIL **Evidence:** pam.pam_faillock.faillock_conf_settings.even_deny_root is null, and pam_faillock is not enabled. **Recommendation:** Enable pam_faillock (see 5.3.2.2) and set even_deny_root in /etc/security/faillock.conf.
- **Rule ID:** 5.3.3.2.1 **Rule Title:** pam_pwquality arguments: difok >= 2 **Status:** FAIL **Evidence:** pam.pam_pwquality.pwquality_conf_settings.difok is null, and pam_pwquality is not enabled. **Recommendation:** Install libpam-pwquality (see 5.3.1.3), enable pam_pwquality (see 5.3.2.3), and set difok = 2 or greater in /etc/security/pwquality.conf.
- **Rule ID:** 5.3.3.2.2 **Rule Title:** pam_pwquality arguments: minlen >= 14 **Status:** FAIL **Evidence:** pam.pam_pwquality.pwquality_conf_settings.minlen is null, and pam_pwquality is not enabled. **Recommendation:** Install libpam-pwquality (see 5.3.1.3), enable pam_pwquality (see 5.3.2.3), and set minlen = 14 or greater in /etc/security/pwquality.conf.
- **Rule ID:** 5.3.3.2.3 **Rule Title:** (Manual) Password complexity (dcredit, ucredit, lcredit, ocredit) **Status:** INFORMATIONAL **Evidence:**

pam.pam_pwquality.pwquality_conf_settings shows dcredit: null, ucredit: null, lcredit: null, ocredit: null. **Recommendation:** N/A (Manual review required after pam_pwquality is enabled and configured).

- **Rule ID:** 5.3.3.2.4 **Rule Title:** pam_pwquality arguments: maxrepeat <= 3 **Status:** FAIL **Evidence:** pam.pam_pwquality.pwquality_conf_settings.maxrepeat is null, and pam_pwquality is not enabled. **Recommendation:** Install libpam-pwquality (see 5.3.1.3), enable pam_pwquality (see 5.3.2.3), and set maxrepeat = 3 or less in /etc/security/pwquality.conf.
- **Rule ID:** 5.3.3.2.5 **Rule Title:** pam_pwquality arguments: maxsequence <= 3 **Status:** FAIL **Evidence:** pam.pam_pwquality.pwquality_conf_settings.maxsequence is null, and pam_pwquality is not enabled. **Recommendation:** Install libpam-pwquality (see 5.3.1.3), enable pam_pwquality (see 5.3.2.3), and set maxsequence = 3 or less in /etc/security/pwquality.conf.
- **Rule ID:** 5.3.3.2.6 **Rule Title:** pam_pwquality arguments: dictcheck = 1 **Status:** FAIL **Evidence:** pam.pam_pwquality.pwquality_conf_settings.dictcheck is null, and pam_pwquality is not enabled. **Recommendation:** Install libpam-pwquality (see 5.3.1.3), enable pam_pwquality (see 5.3.2.3), and set dictcheck = 1 in /etc/security/pwquality.conf.
- **Rule ID:** 5.3.3.2.7 **Rule Title:** pam_pwquality arguments: enforcing = 1 **Status:** FAIL **Evidence:** pam.pam_pwquality.pwquality_conf_settings.enforcing is null, and pam_pwquality is not enabled. **Recommendation:** Install libpam-pwquality (see 5.3.1.3), enable pam_pwquality (see 5.3.2.3), and set enforcing = 1 in /etc/security/pwquality.conf.
- **Rule ID:** 5.3.3.2.8 **Rule Title:** pam_pwquality arguments: enforce_for_root present **Status:** FAIL **Evidence:** pam.pam_pwquality.pwquality_conf_settings.enforce_for_root is null, and pam_pwquality is not enabled. **Recommendation:** Install libpam-pwquality (see 5.3.1.3), enable pam_pwquality (see 5.3.2.3), and set enforce_for_root in /etc/security/pwquality.conf.
- **Rule ID:** 5.3.3.3.1 **Rule Title:** pam_pwhistory arguments: remember >= 24 **Status:** FAIL **Evidence:** pam.pam_pwhistory.pwhistory_inline_pam_settings.remember is null, and pam_pwhistory is not enabled. **Recommendation:** Enable pam_pwhistory (see 5.3.2.4) and set remember = 24 or greater in its PAM configuration (e.g., in /etc/pam.d/common-password).
- **Rule ID:** 5.3.3.3.2 **Rule Title:** pam_pwhistory arguments: enforce_for_root present **Status:** FAIL **Evidence:** pam.pam_pwhistory.pwhistory_inline_pam_settings.enforce_for_root is null, and pam_pwhistory is not enabled. **Recommendation:** Enable pam_pwhistory (see 5.3.2.4) and add enforce_for_root to its PAM configuration (e.g., in /etc/pam.d/common-password).
- **Rule ID:** 5.3.3.3.3 **Rule Title:** pam_pwhistory arguments: use_authok present **Status:** FAIL **Evidence:** pam.pam_pwhistory.pwhistory_inline_pam_settings.use_authok is null, and pam_pwhistory is not enabled. **Recommendation:** Enable pam_pwhistory (see 5.3.2.4) and add use_authok to its PAM configuration (e.g., in /etc/pam.d/

common-password).

- **Rule ID:** 5.3.3.4.1 **Rule Title:** pam_unix arguments: nullok absent **Status:** FAIL
Evidence: pam.pam_unix.pam_unix_flag_presence.nullok is "present" in /etc/pam.d/common-auth. **Recommendation:** Remove nullok from the pam_unix.so line in /etc/pam.d/common-auth to prevent authentication with empty passwords.
- **Rule ID:** 5.3.3.4.2 **Rule Title:** pam_unix arguments: remember absent **Status:** PASS
Evidence: pam.pam_unix.pam_unix_flag_presence.remember is null.
Recommendation: N/A
- **Rule ID:** 5.3.3.4.3 **Rule Title:** pam_unix arguments: Strong hashing **Status:** PASS
Evidence: pam.pam_unix.pam_unix_flag_presence.yescrypt is "present".
Recommendation: N/A
- **Rule ID:** 5.3.3.4.4 **Rule Title:** pam_unix arguments: use_authtok present (in password stack) **Status:** FAIL **Evidence:**
pam.pam_unix.pam_unix_flag_presence.use_authtok is null.
Recommendation: Add use_authtok to the pam_unix.so line in /etc/pam.d/common-password to ensure the new password is used for all modules in the password stack.

SUID/SGID Triage Report

```
{
  "findings": [],
  "summary": {
    "suspicious_found": 0,
    "message": "No unexpected SUID/SGID files detected."
  }
}
```